

Smart Document Similarity and Plagiarism Detection System

Data Structures & Algorithms — Project Presentation

Review 1

SECTION A7

GROUP G-9

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Similarity:
87%

Introduction

The **Smart Document Similarity and Plagiarism Detection System** compares documents to identify similar or copied content efficiently. It processes uploaded documents, extracts and preprocesses text, and generates a similarity report — reducing manual effort while demonstrating the practical application of Data Structures and Algorithms.

Planned DSA Concepts



**Tokenization & Text
Preprocessing**



Hash Tables (HashMap)



AVL Trees



Searching & Sorting



Dynamic Programming

Planned Team Roles

Each team member is assigned a focused area of responsibility for the planned implementation.



Vidya Keerthi Chitturi

- Text preprocessing
- Tokenization
- String processing

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Shaik Asha Parveen

- AVL Tree implementation
- Searching algorithms
- Sorting algorithms



Madi Amulya

- Hash Tables (HashMap)
- Word frequency analysis

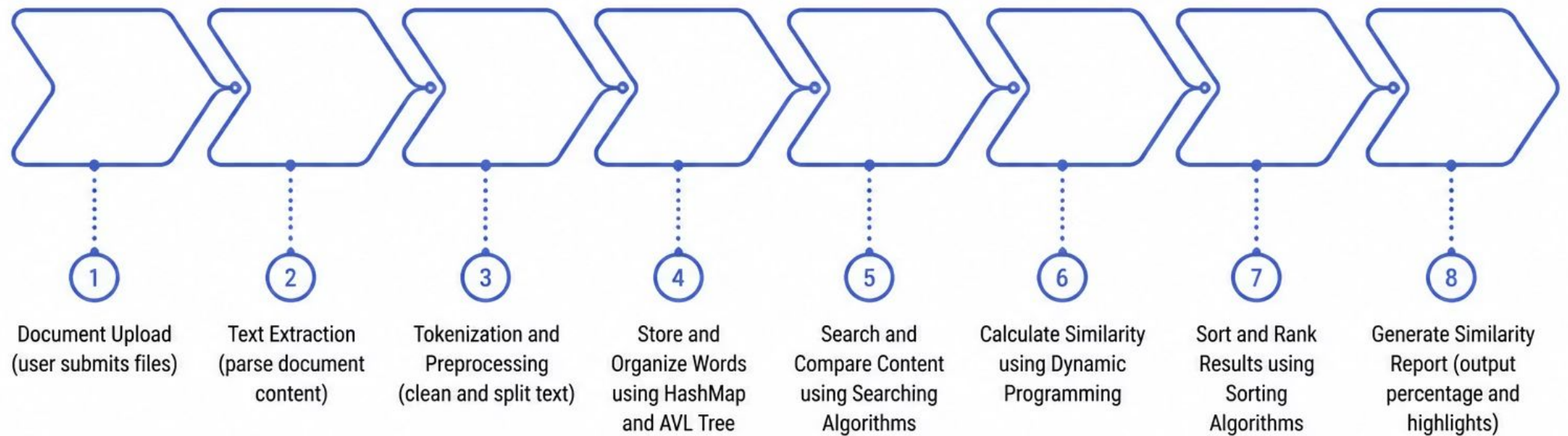


Badam Hasika

- Dynamic Programming
- Similarity calculation
- Testing & report generation

Planned System Workflow

The end-to-end pipeline applies DSA concepts at each stage — from raw document input to a structured similarity report.

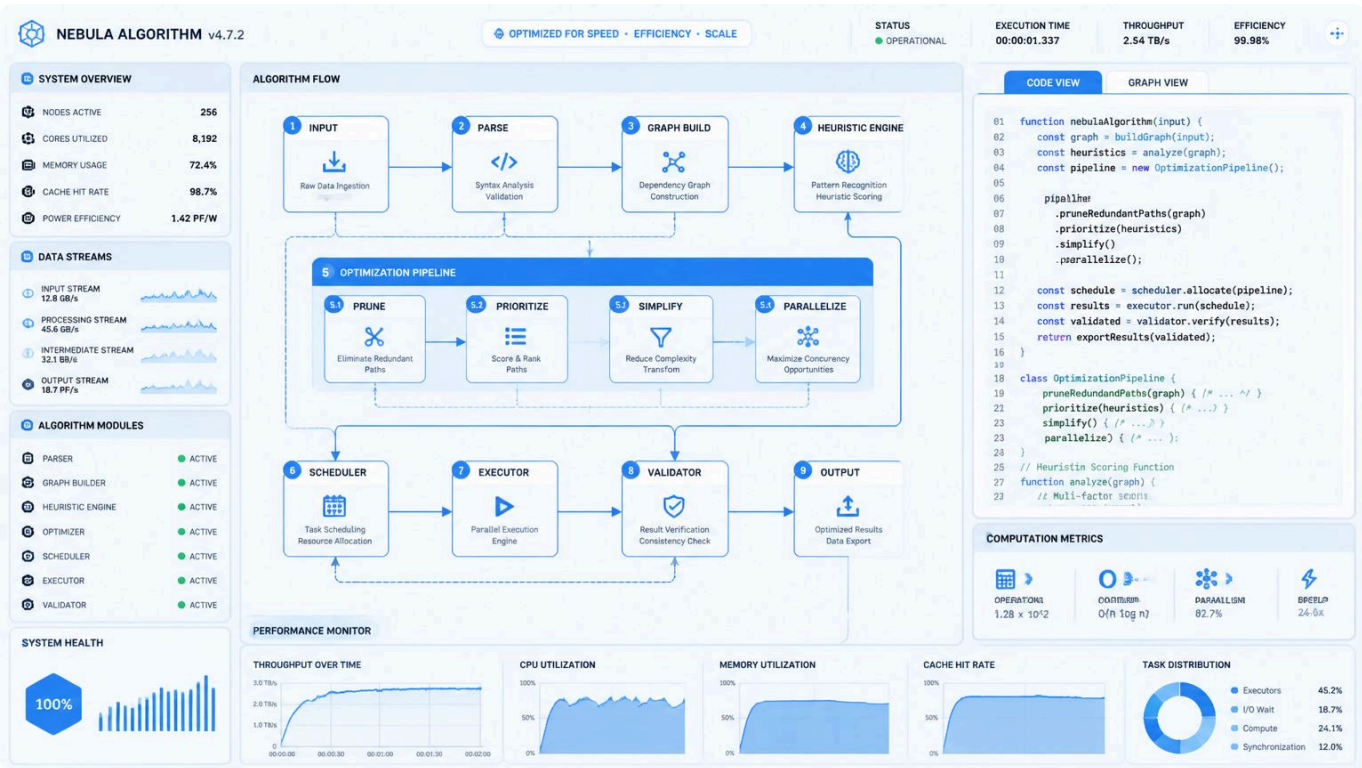


Expected Outcomes

The project aims to achieve the following outcomes through planned DSA-based implementation:

- Detect **exact and near-duplicate content** between documents.
- Calculate and display a **similarity percentage** for each comparison.
- Identify and **highlight matching text** segments within documents.
- Generate a **clear, structured similarity report** for review.
- **Reduce time and effort** required for manual plagiarism checking.
- Demonstrate the **practical use of DSA** in document analysis.

Advantages



Time-Saving Automation

Automates document comparison, significantly reducing manual effort.

Efficient Word Lookup

HashMap enables $O(1)$ average-time storage and retrieval of words.

Balanced Search Structure

AVL Trees maintain $O(\log n)$ search even as data grows.

Quantifiable Results

Provides similarity scores and ranked comparison output.

Scalable Design

Architecture can be extended to compare multiple documents.

Applications

The system is designed for any context where document originality and content overlap must be verified.



Educational Institutions

Checking assignments, projects, and student reports for originality.



Publishing Companies

Detecting duplicate content before manuscripts go to press.



Corporate Organizations

Comparing internal reports and technical documentation.



Research Organizations

Verifying the originality of research papers and publications.



Online Learning Platforms

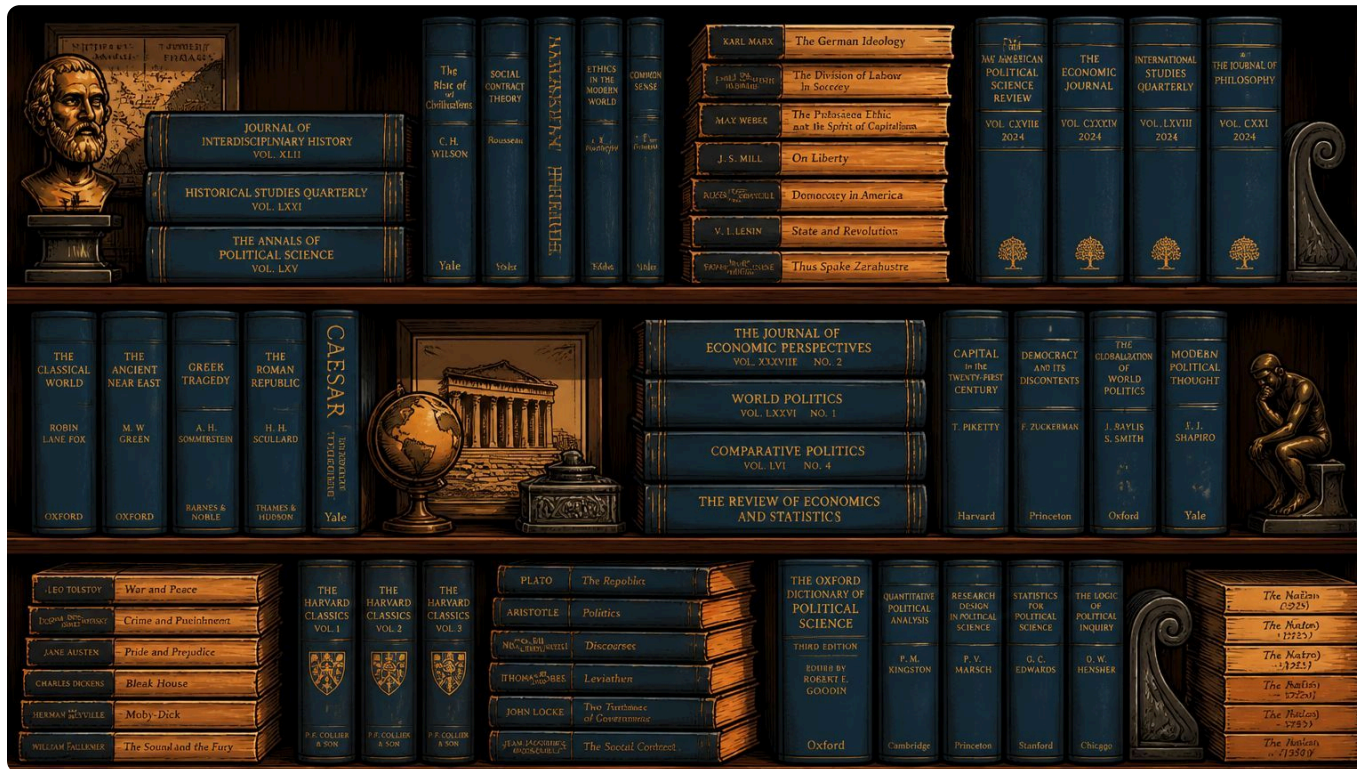
Reviewing submitted coursework across large student populations.



Content Creation Platforms

Ensuring uniqueness of articles, blogs, and web content.

References



Academic Sources

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Helgesson, G., & Eriksson, S. (2015). Plagiarism in Research. *Medicine, Health Care and Philosophy*, 18(1), 91–101. <https://doi.org/10.1007/s11019-014-9583-8>